

Advanced Imaging Technologies in Systems Biology and Biomedical Research

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Contact information

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RSM 3.14

Contact options: Use Ed discussion board



Ed Discussion board 

Advanced Imaging Technologies for Biomedical Research

☐ Which microscope to use?

☐ What optical imaging probe to employ?

☐ Which assay/sensor to apply?

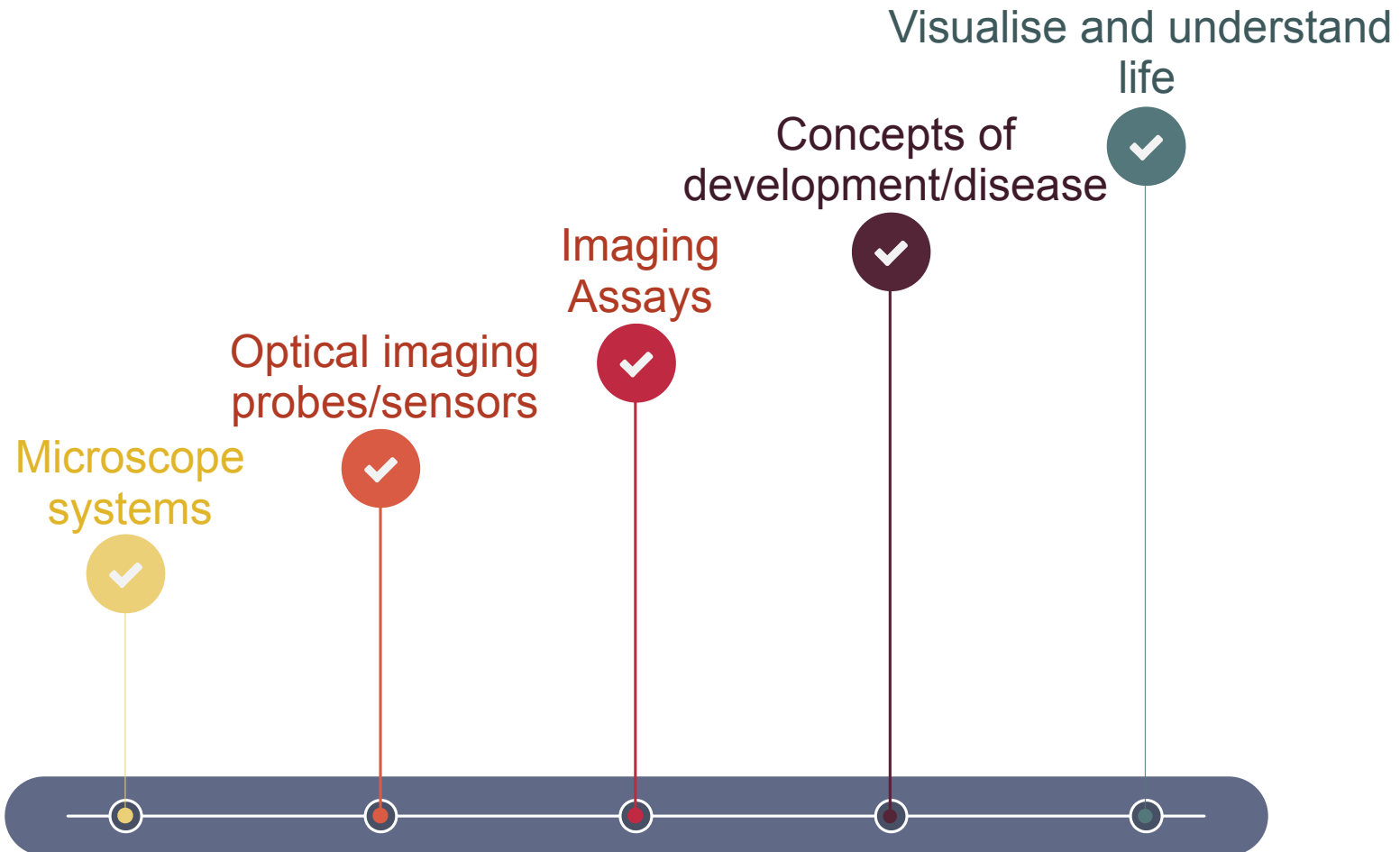
☐ Which model organism to investigate?

Learning outcomes

Appreciate the use of advanced imaging technologies to visualise and/or quantify complex biological events.

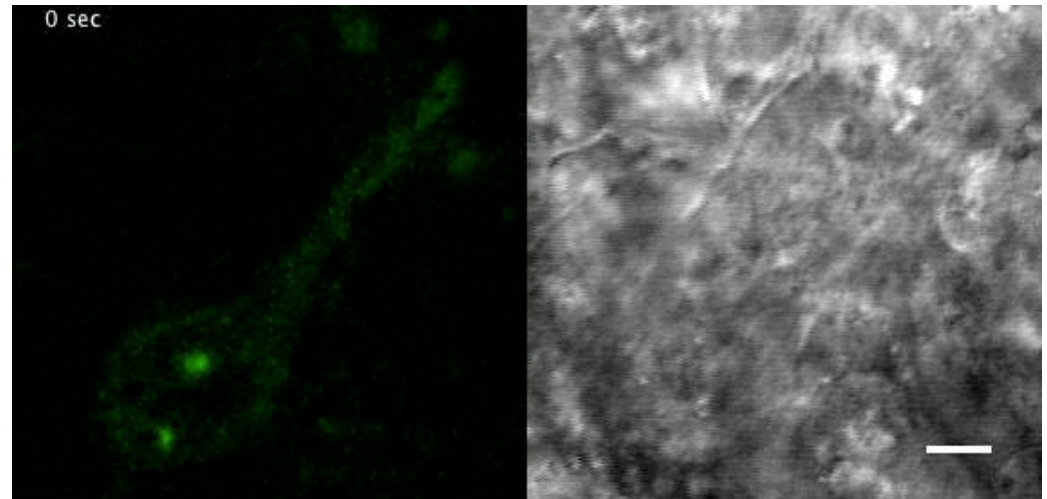
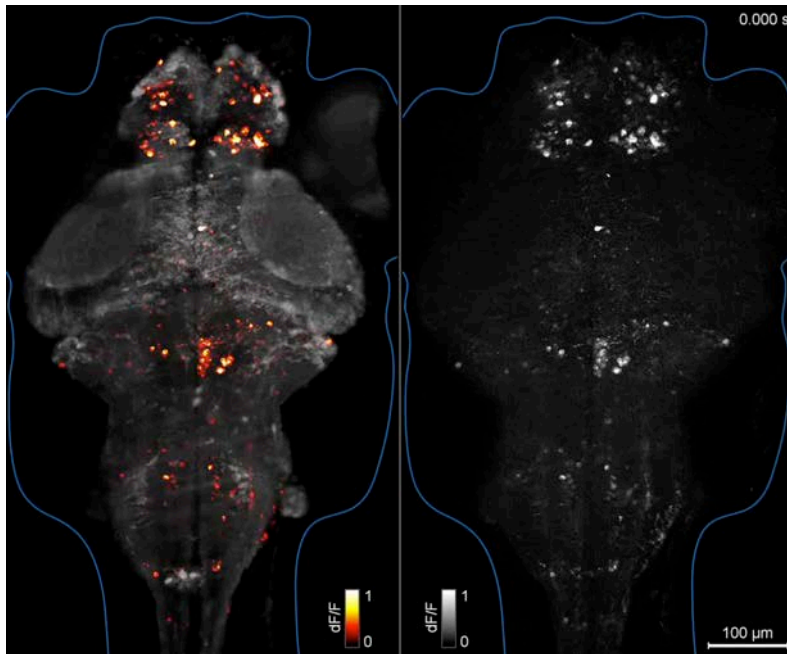
Evaluate experimental approaches and test hypotheses by designing imaging experiments.

Seeing is believing - timeline of lecture content



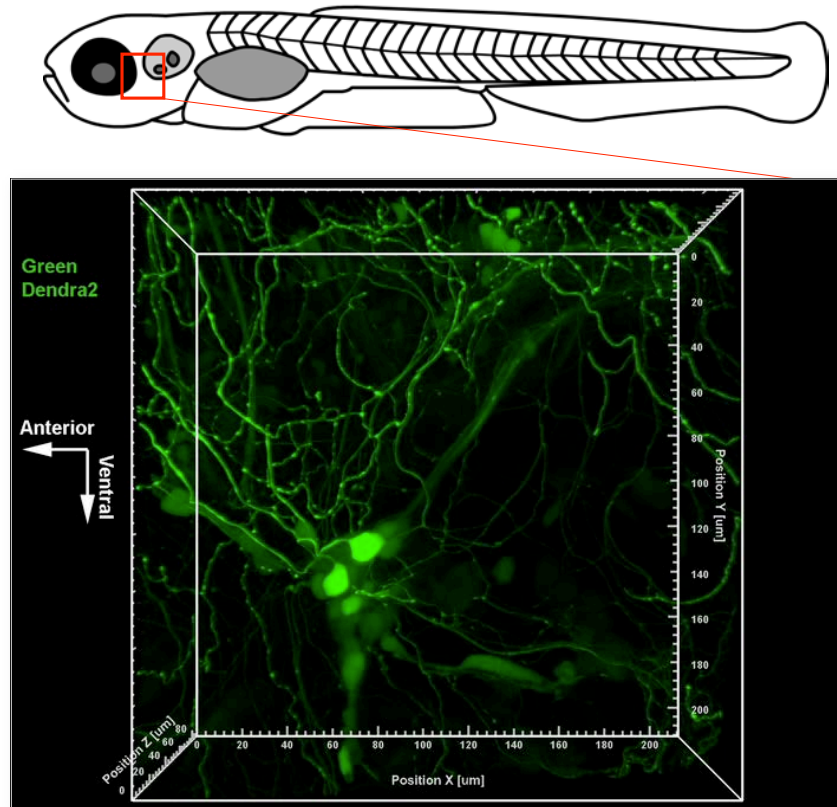
The Challenge

How to visualise activities in brain or heart?



The Challenge

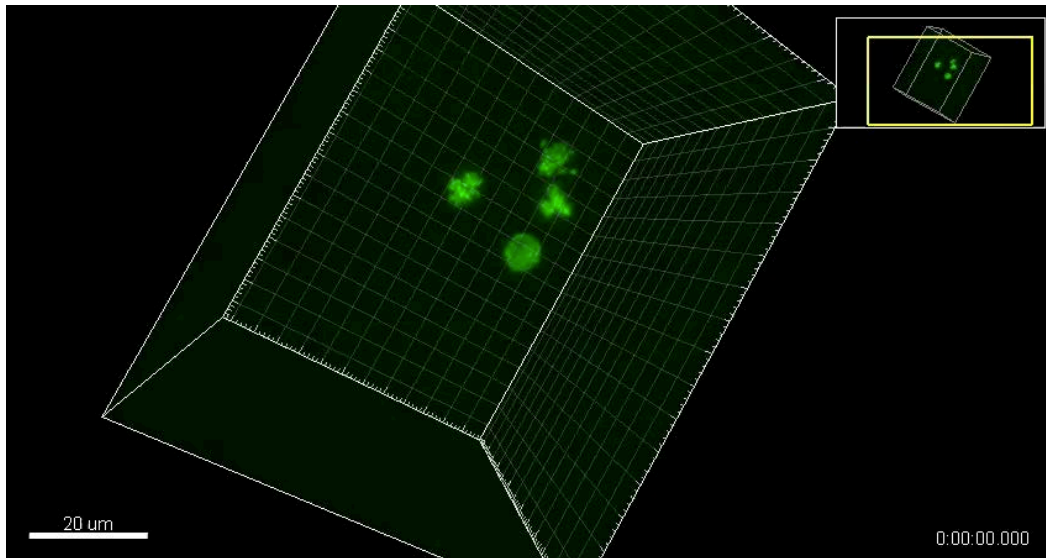
How to unravel complex structural information in the brain?



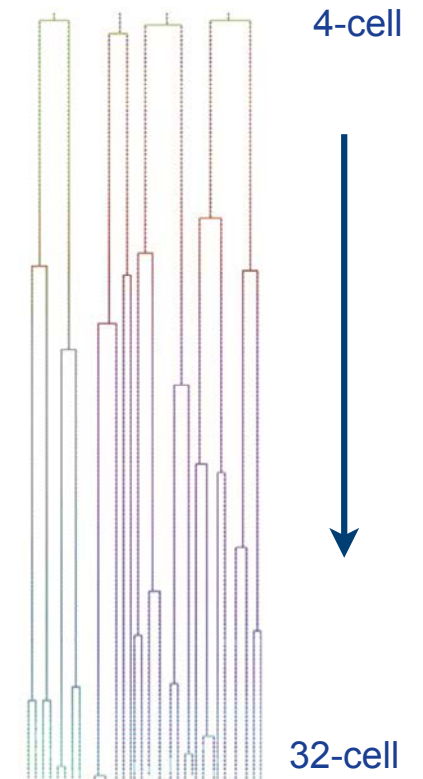
The Challenge

How to display complex dynamics in embryos/organoids

3 Days of Mouse Embryo Development

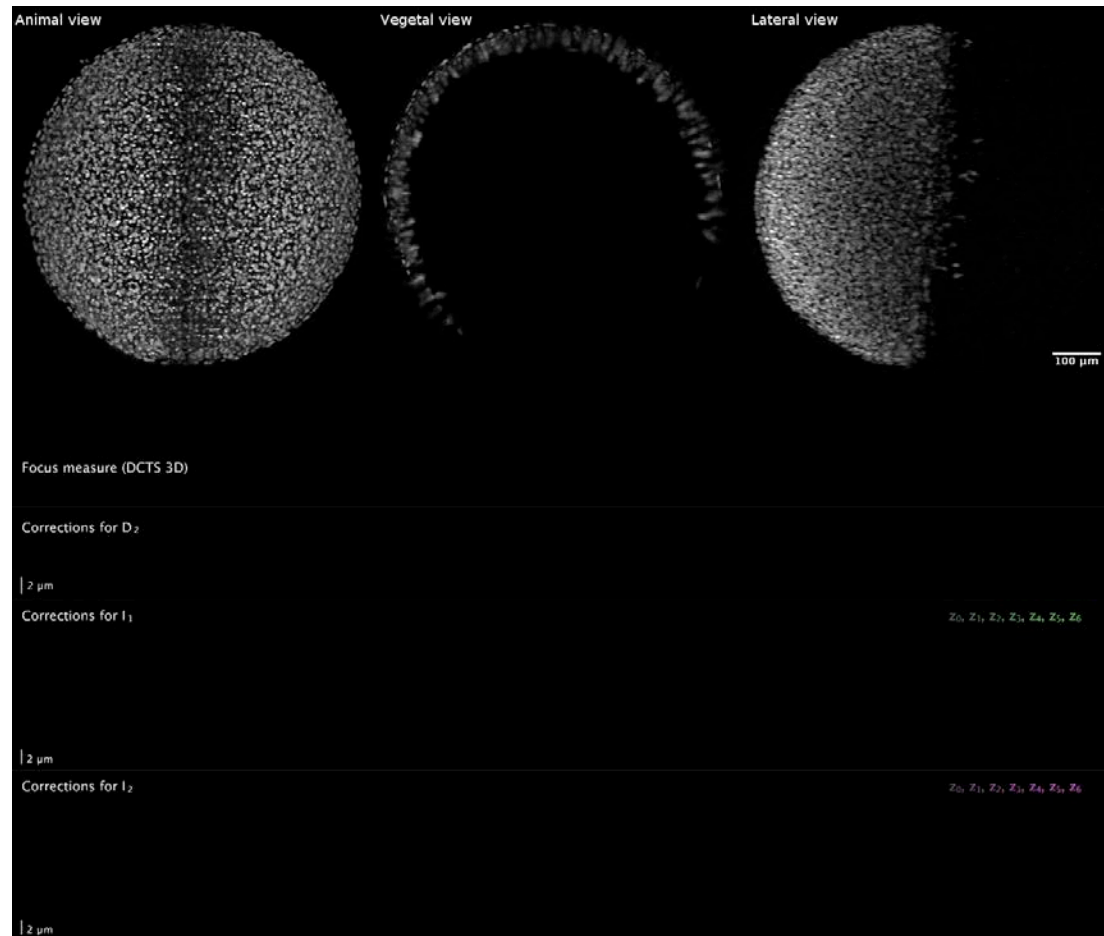


Precise
Lineage Tree



The Challenge

How to capture development and disease as it unfolds?



The Challenge

Can colourful be useful for cancer evaluation and therapy evaluation?



Organisational matters

- 9 lectures
(slides to be uploaded to blackboard)
 - Questions to be placed on Ed discussion board
(Will thrive to address/endorse them within a week)
 - Worksheet will be uploaded before study group
-

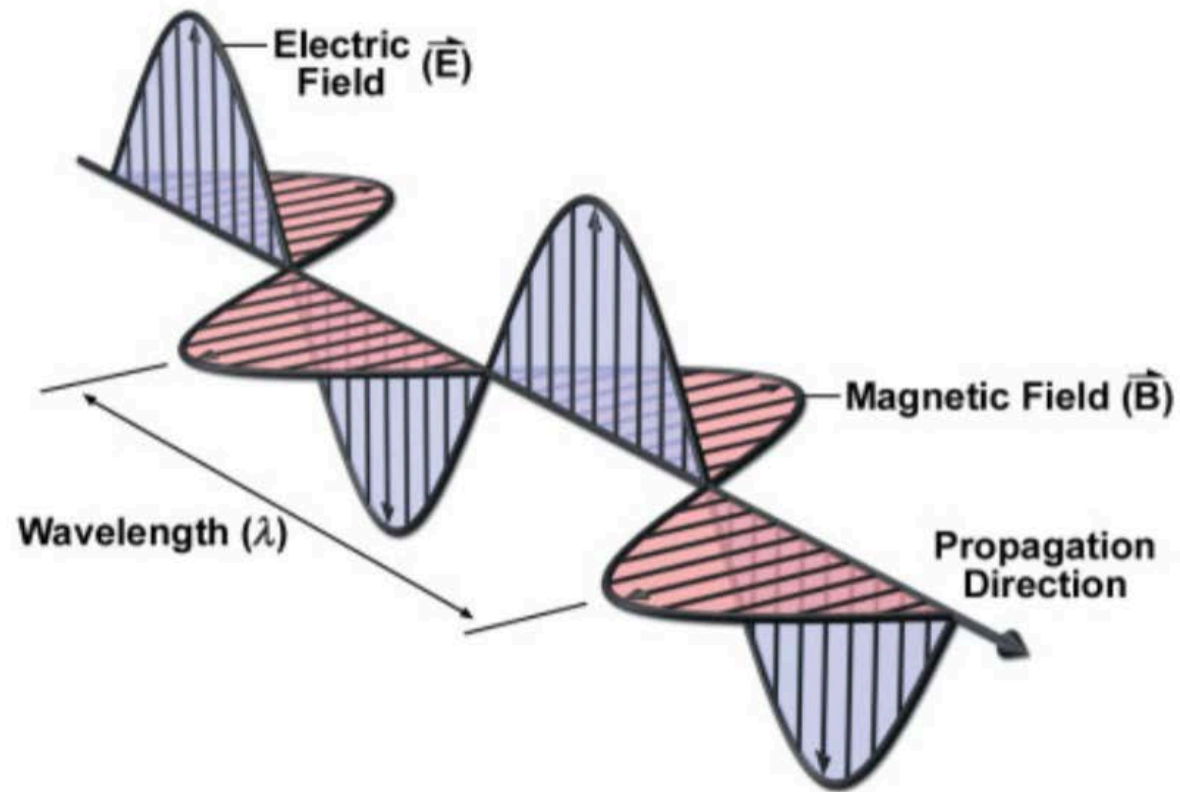
What we will cover

- What is light and colour?
 - How does a light microscope work?
-

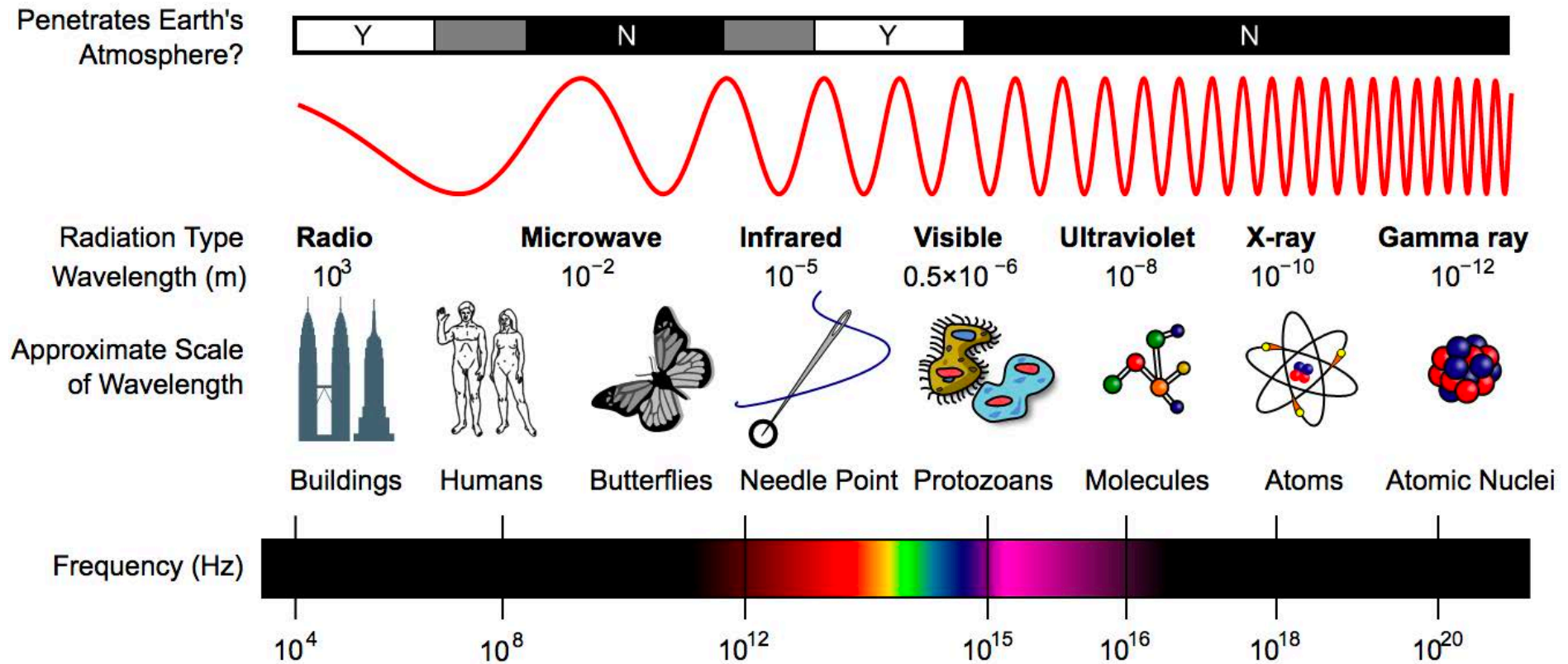
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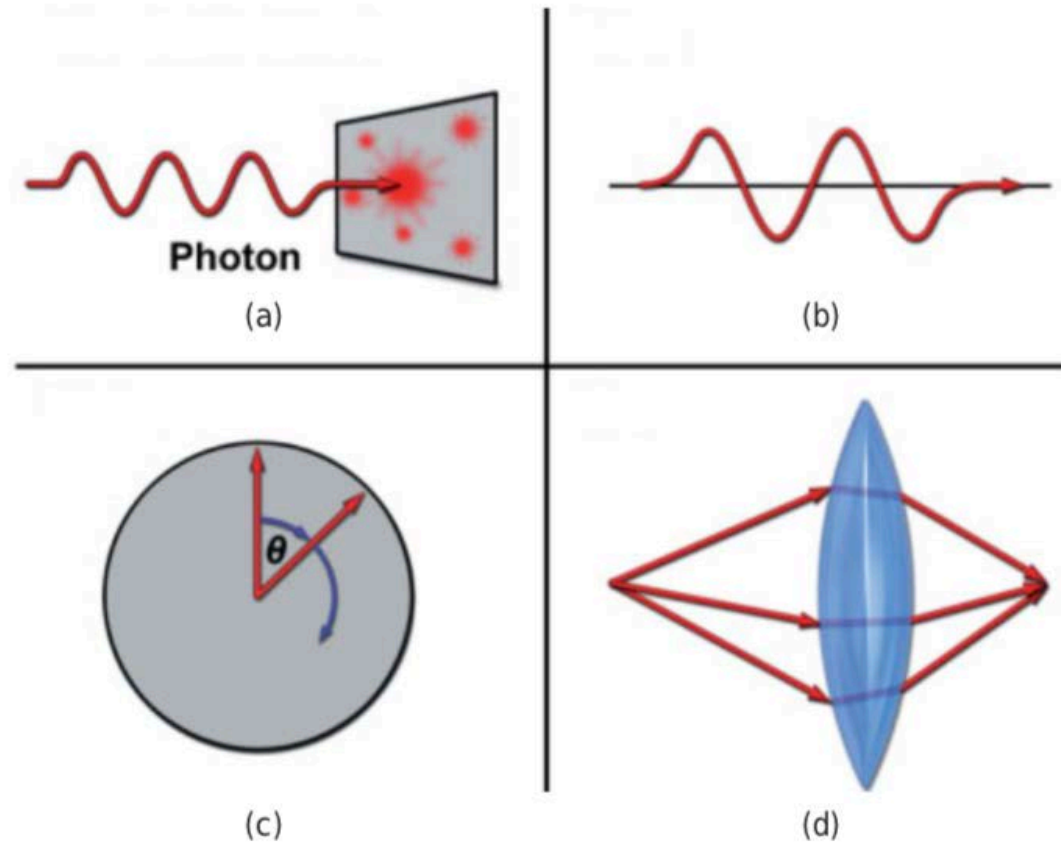
Properties of light: electromagnetic wave



Properties of light: electromagnetic spectrum



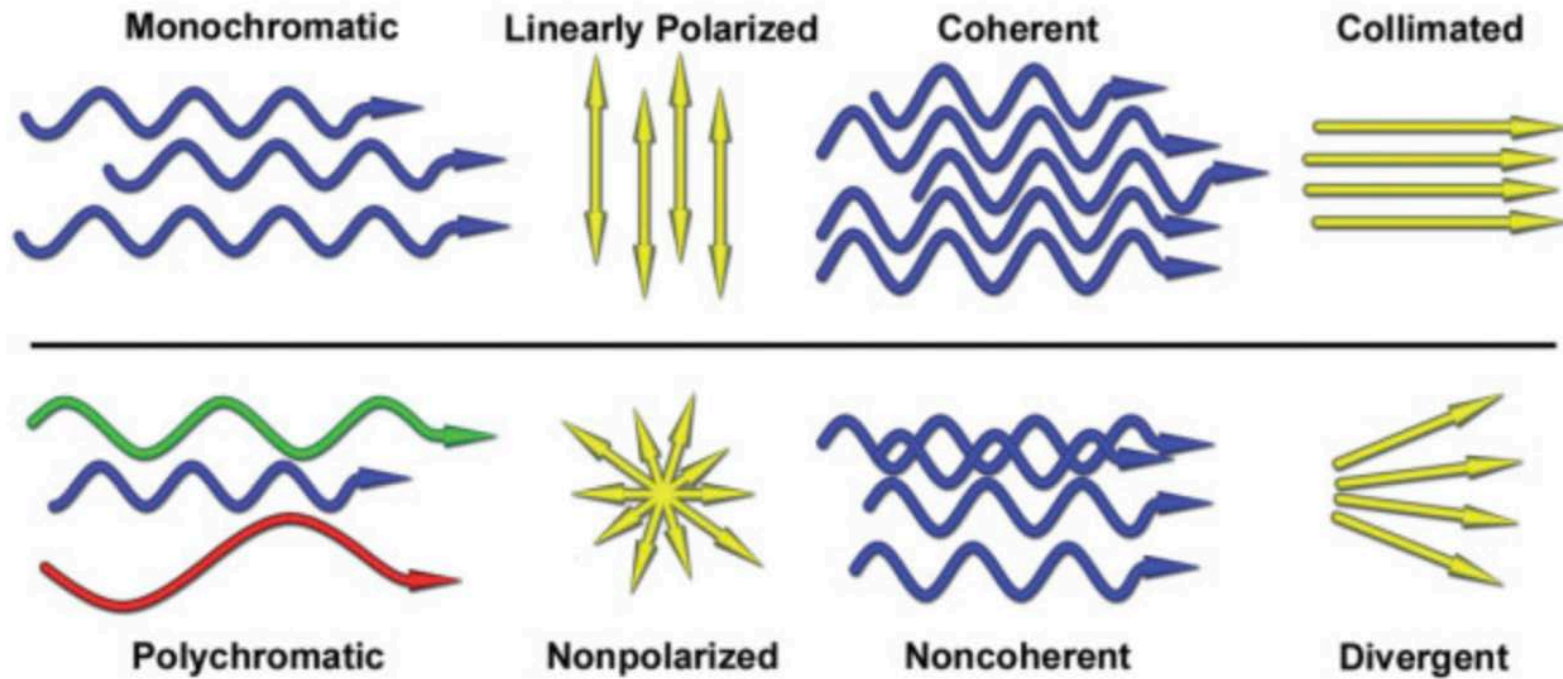
The dual particle- and wave-like nature of light



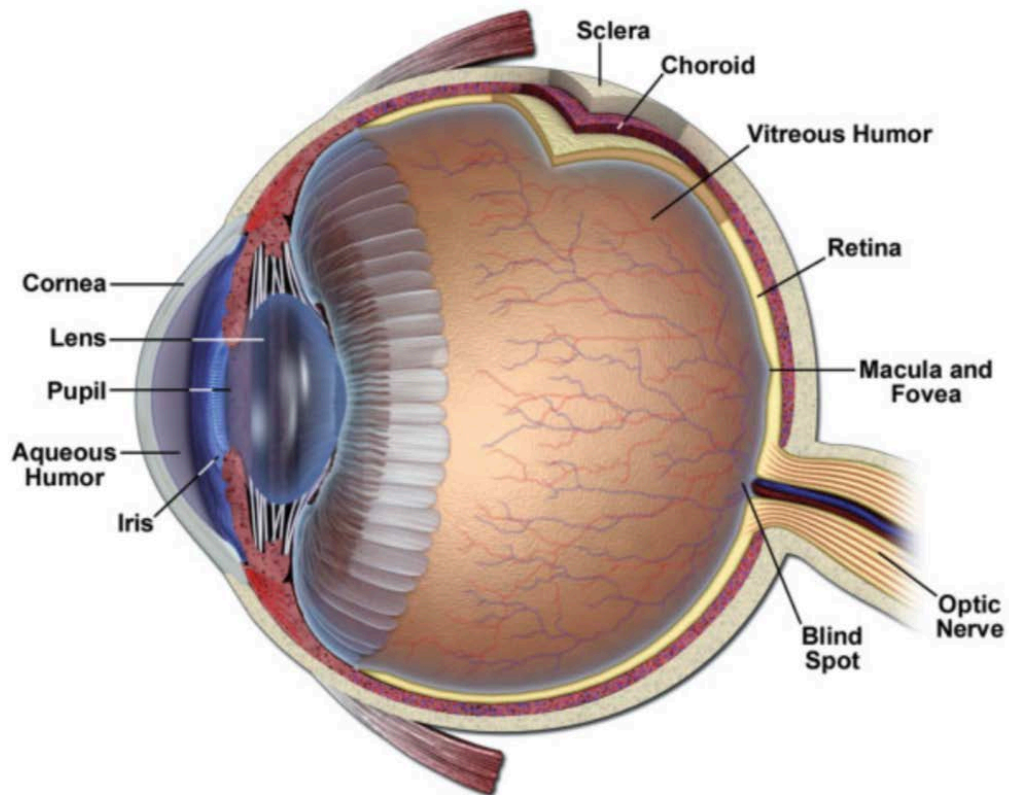
Mathematical relationship of light as particle and wave



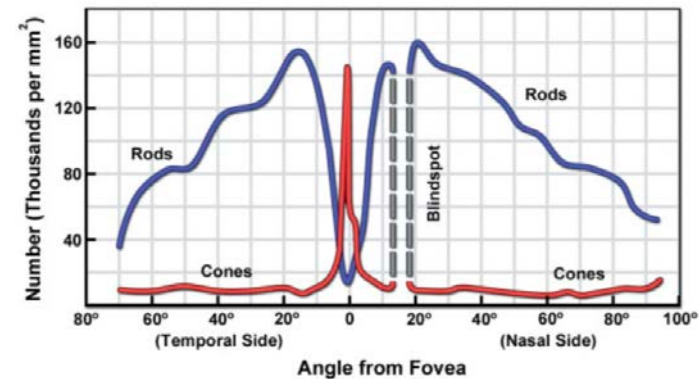
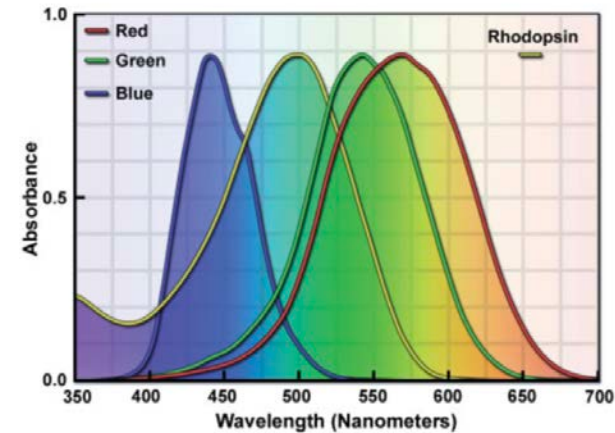
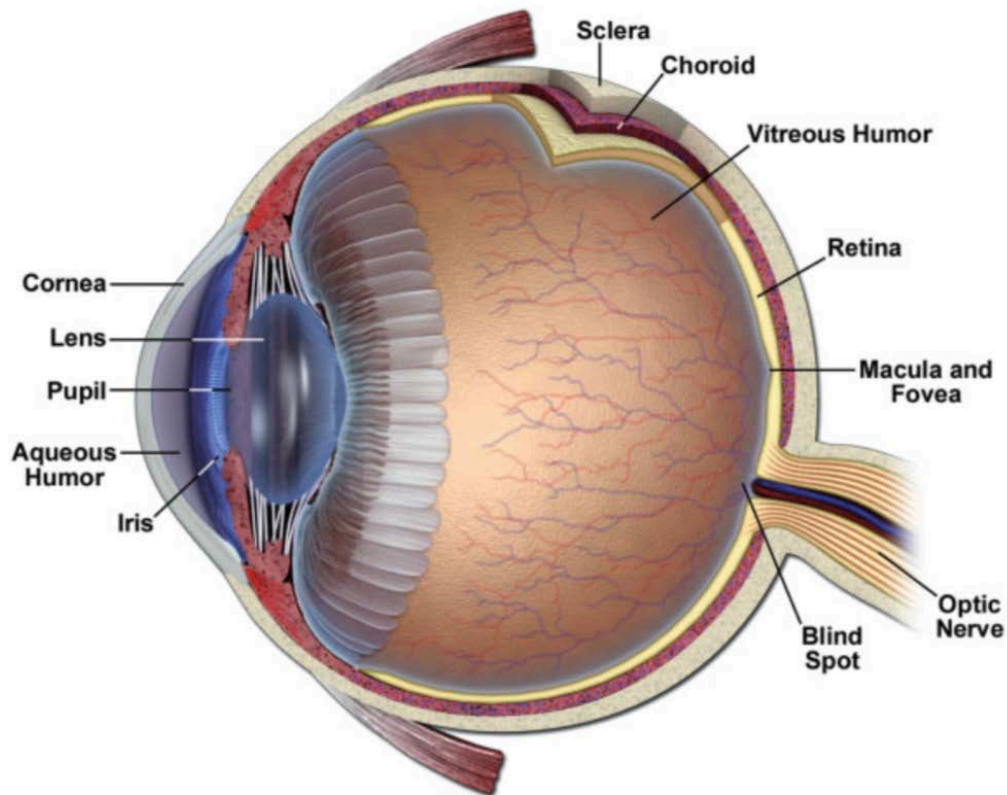
The quality of light



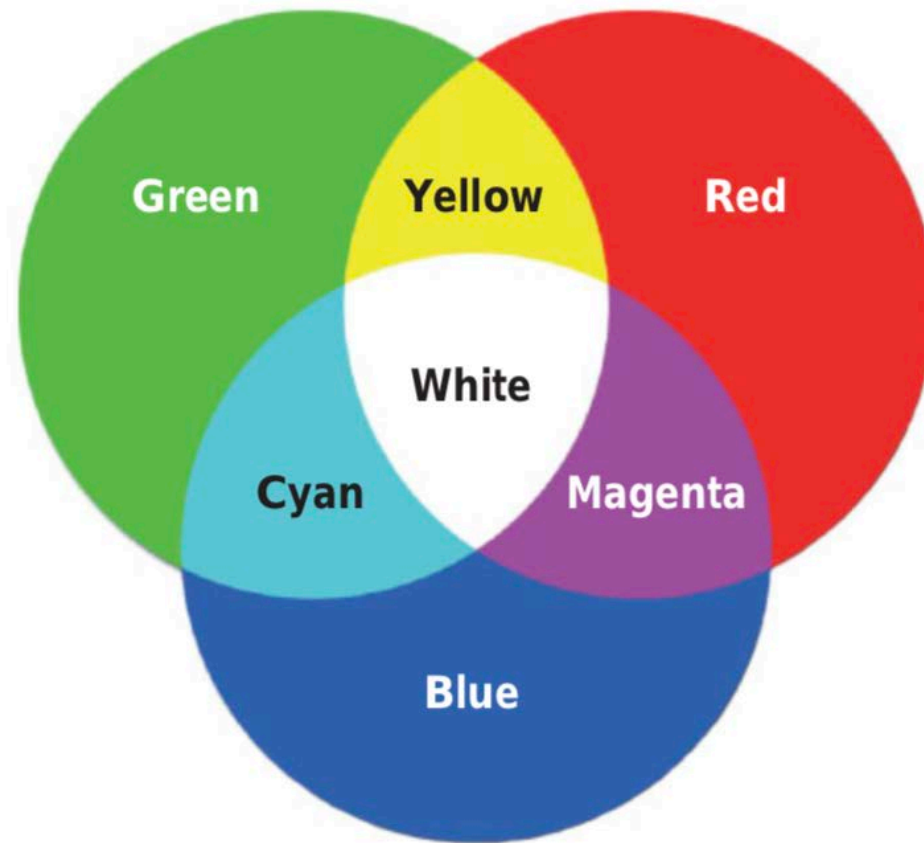
The structure of the human eye



Light perceived by the human eye: Night/Day vision and colour



Colour vision: tristimulus system



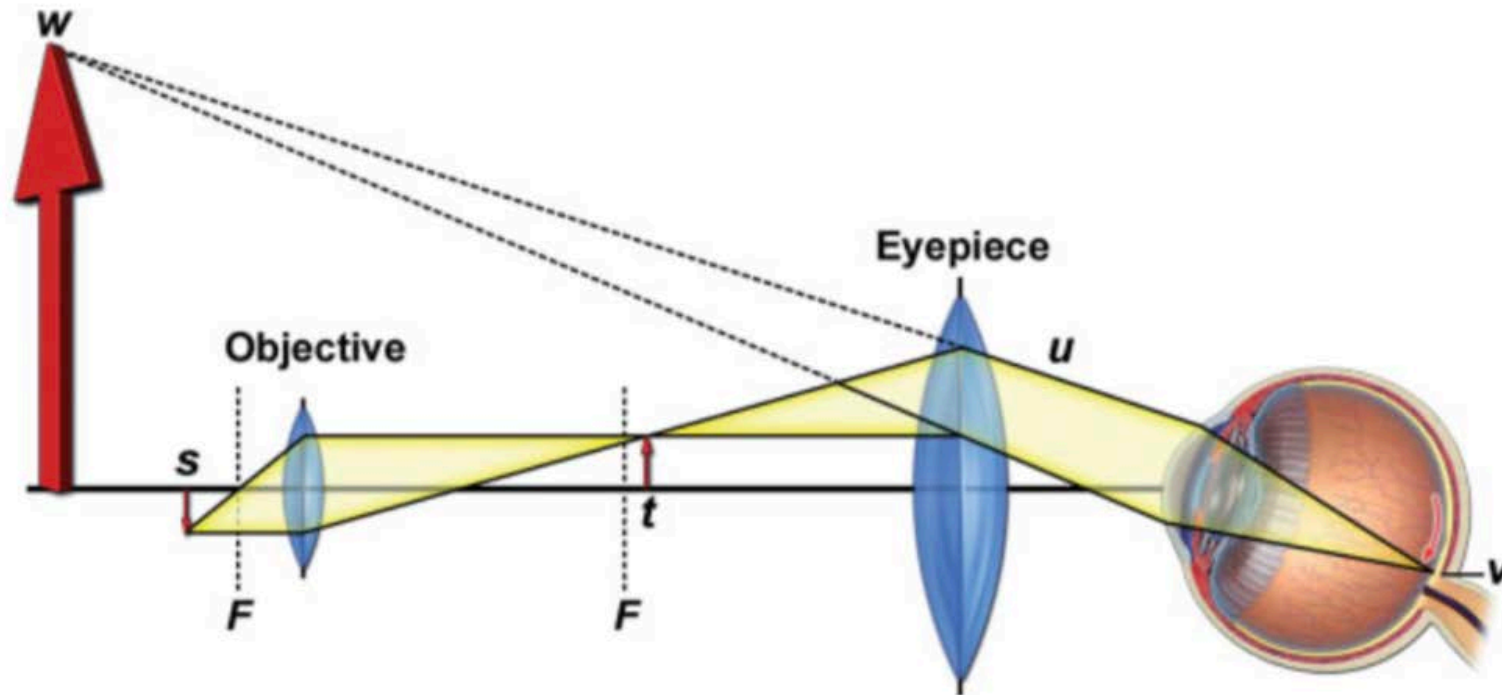
What we will cover

- What is light and colour?
 - **How does a light microscope work?**
-

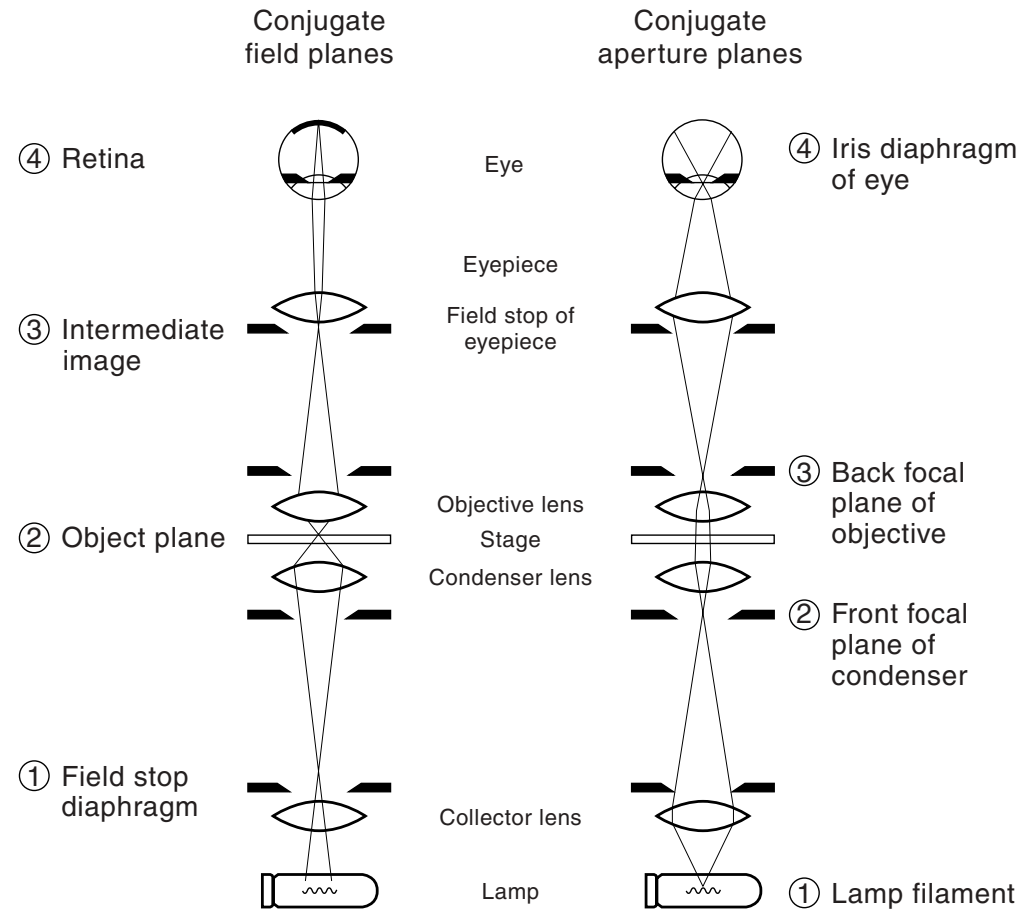
The compound light microscope



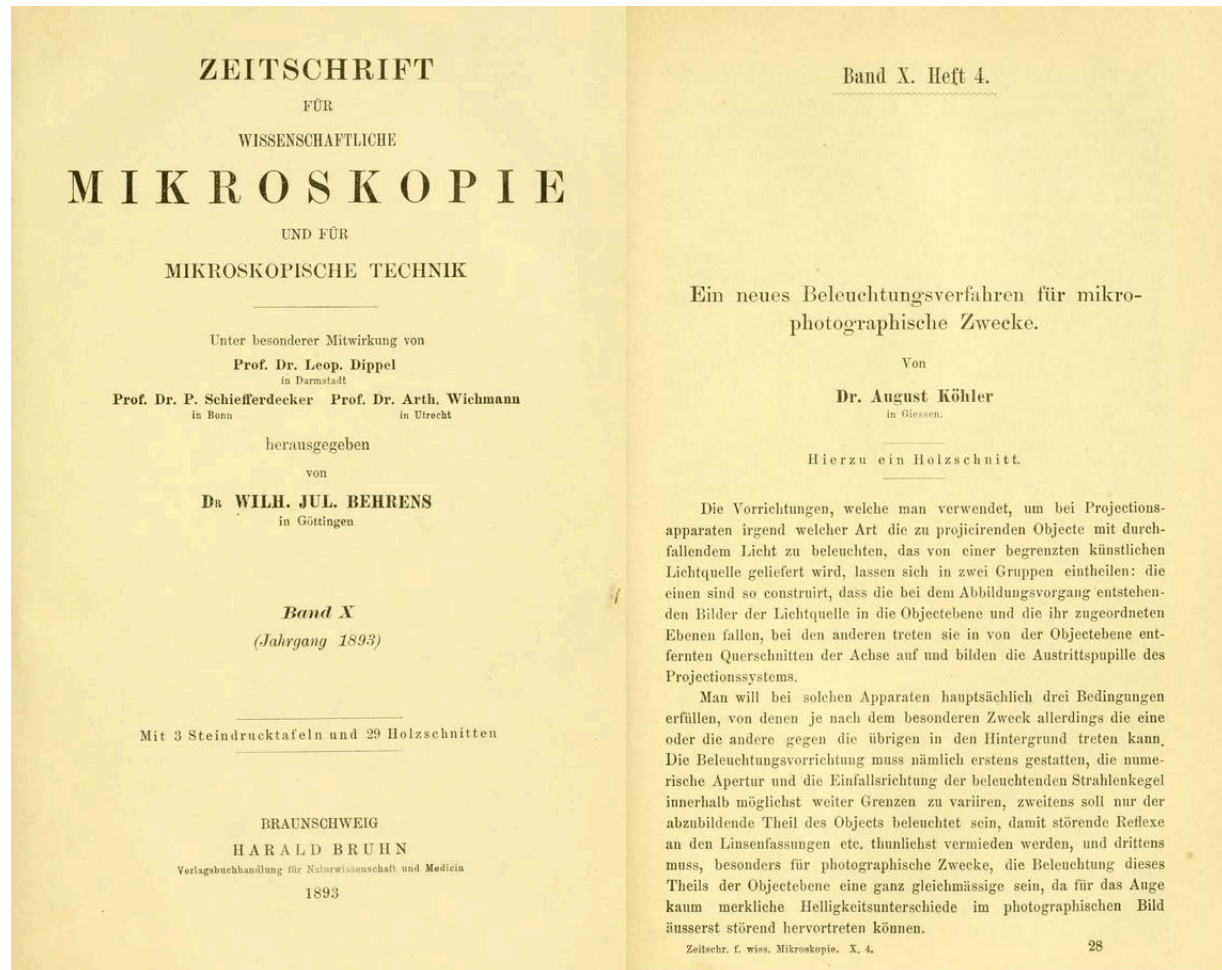
Magnification of an object in the microscope



Field planes and aperture planes

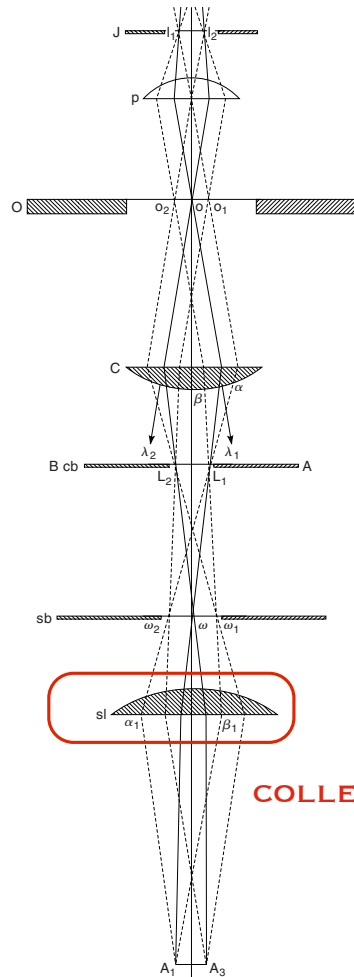


Köhler illumination



Köhler illumination

bright and even illumination



August Köhler
1866-1948



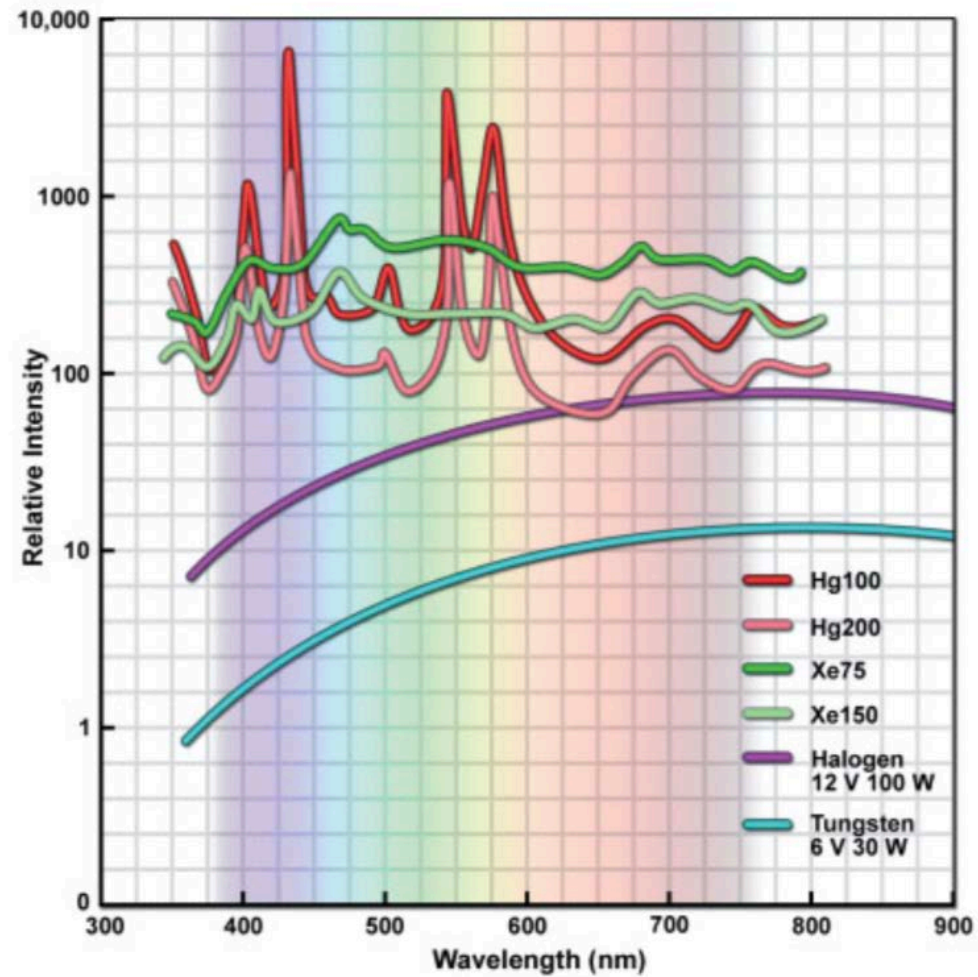
Precaution for handling optical equipment



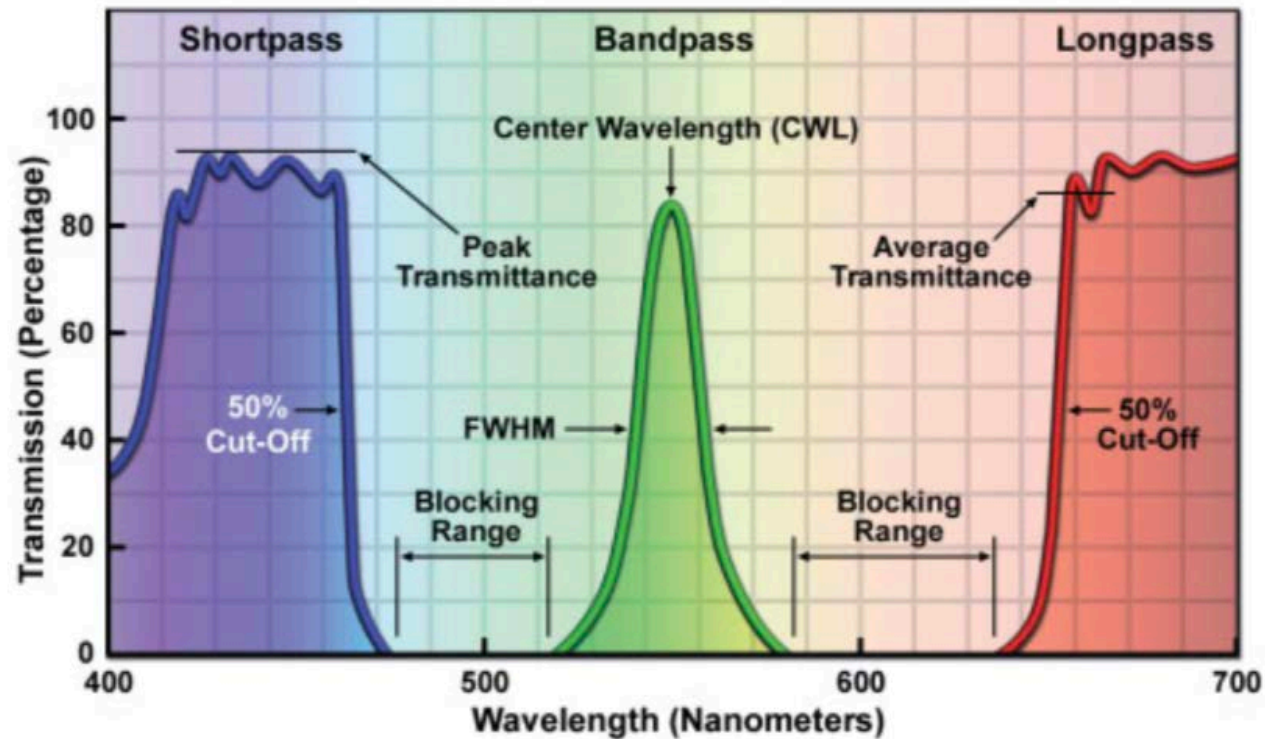
Precaution for handling optical equipment

- Never strain, twist, or drop objectives or other optical components.
 - Never force the focus controls of the objective or condenser, and always watch lens surfaces as they approach the specimen.
 - Never touch optical surfaces.
-

Illuminators



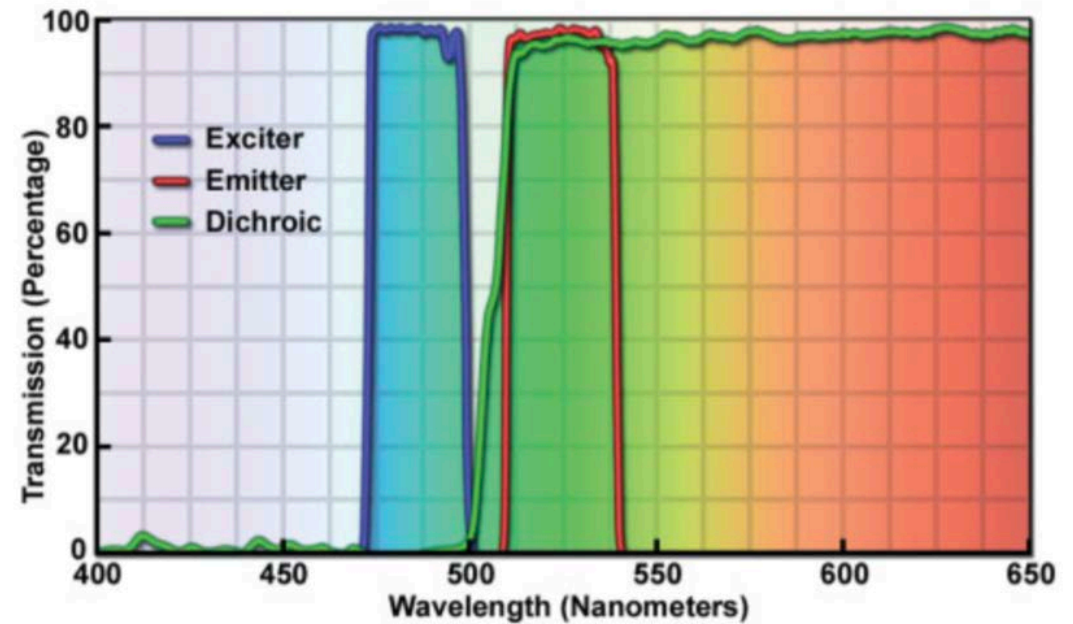
Filters for isolating the wavelength of illumination



Filters for adjusting the wavelength



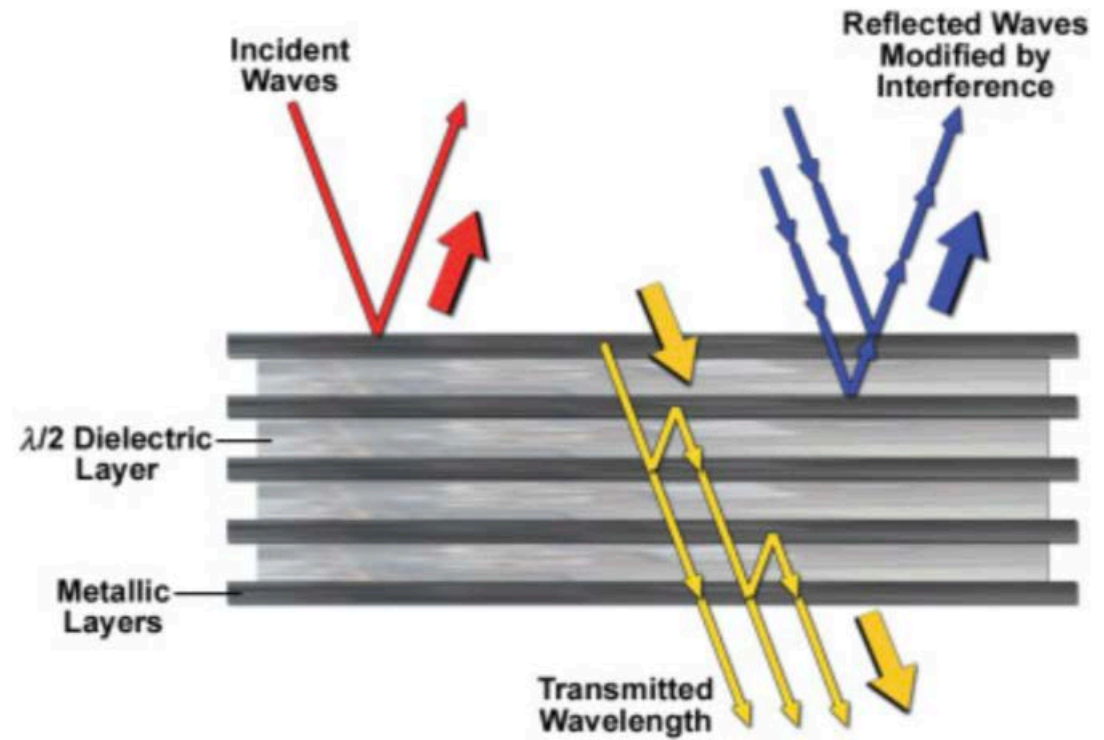
Interference filters



Interference filter selectively transmit waves



Interference filters



Filters for adjusting the intensity



- Neutral Density Filters

$$OD = \log_{10} (1/T)$$

OD = Optical Density

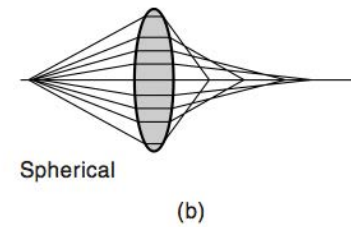
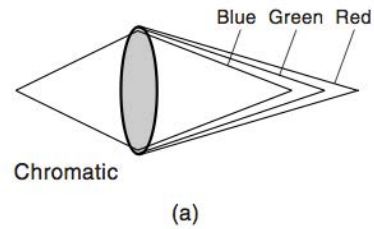
T = Transmittance

The compound light microscope - objectives

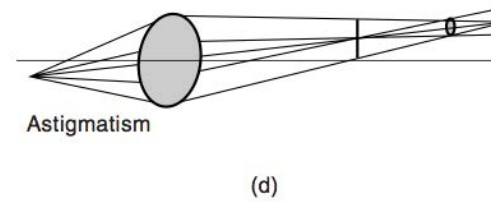
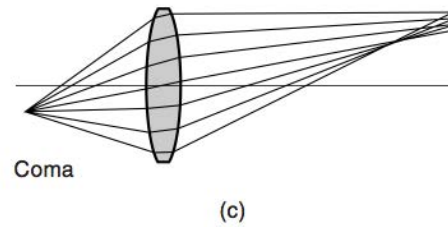


Aberrations of simple lenses

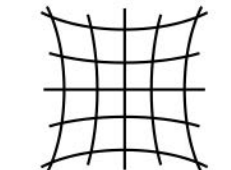
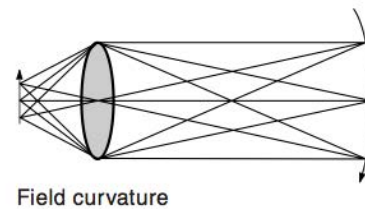
On-axis aberrations



Off-axis aberrations

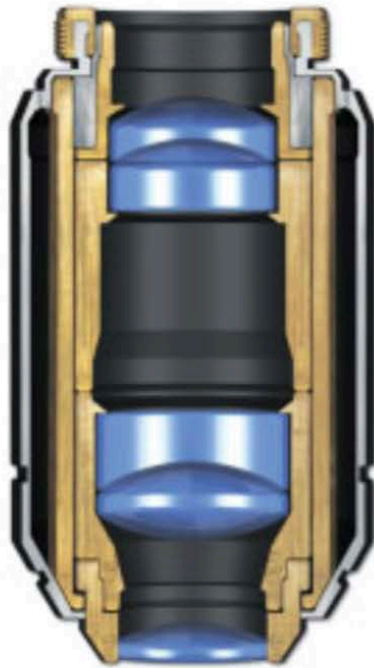


Distortion

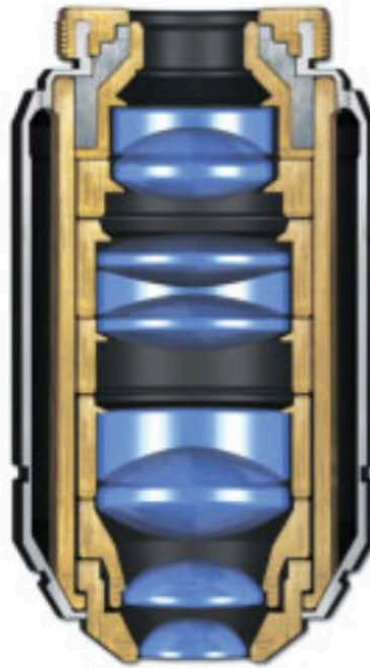


(e)

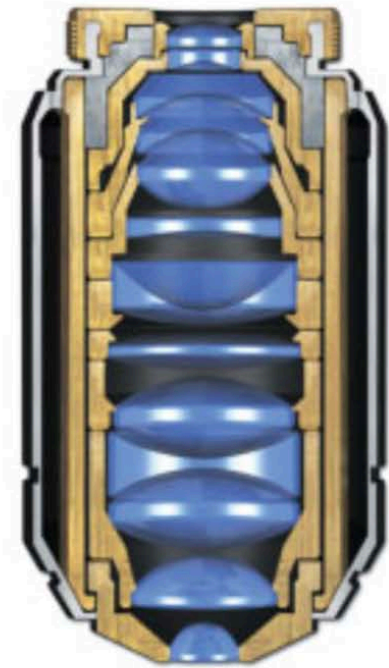
Objective designs



Achromat

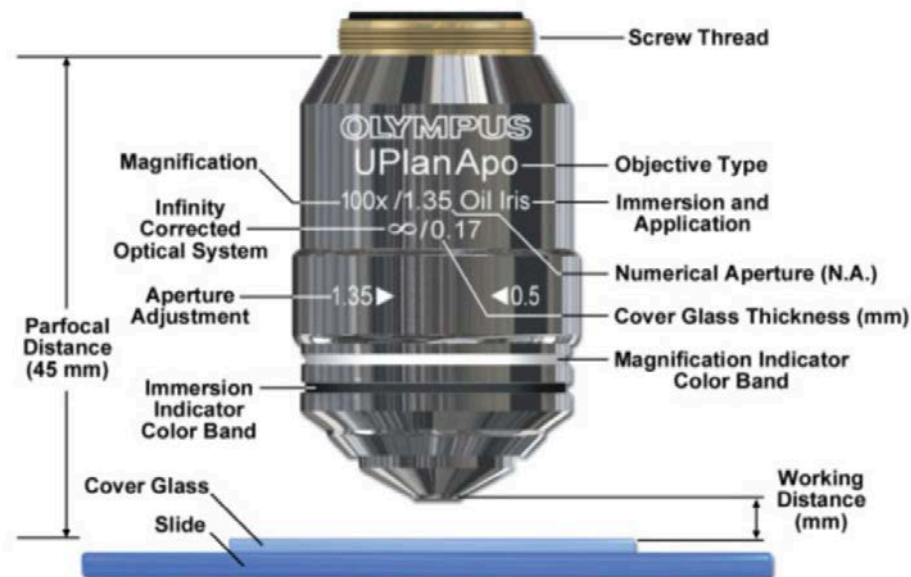


Fluorite



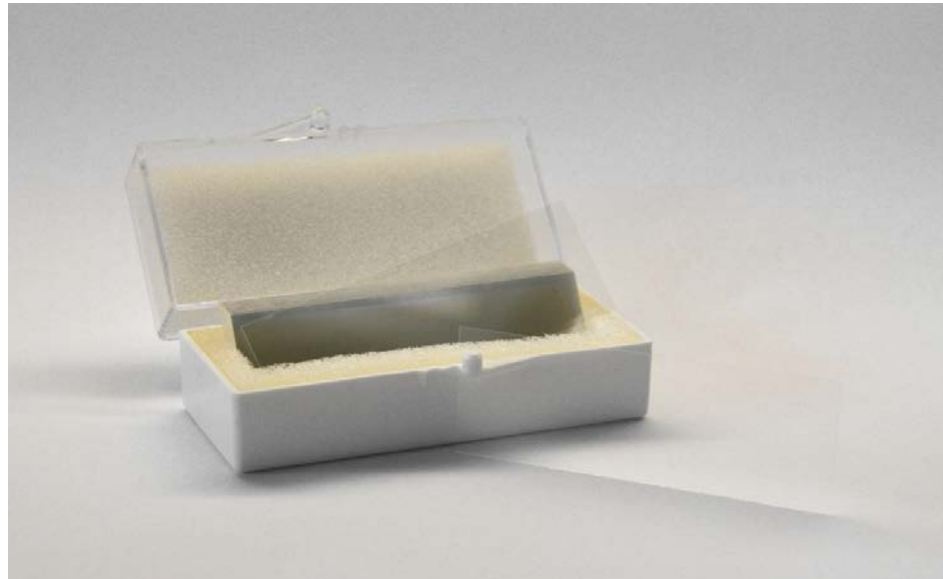
Plan Apochromat

Key for interpreting the markings on the decorative barrel of an objective



Mag.	1×	2×	4×	10×	20×	40×	50×	60×	100×
Code	Black	Gray	Red	Yellow	Green	Light Blue	Dark Blue	White	
Imm. Med.	Oil		Water		Glycerin		Oil/Water/Glycerin		
Code	Black		White		Orange		Red		

Many objectives are designed to be used with coverslips of a certain thickness



Grade Number	Thickness (mm)
0	0.083–0.13
1	0.13–0.16
1.5	0.16–0.19
2	0.19–0.25

Thank you!